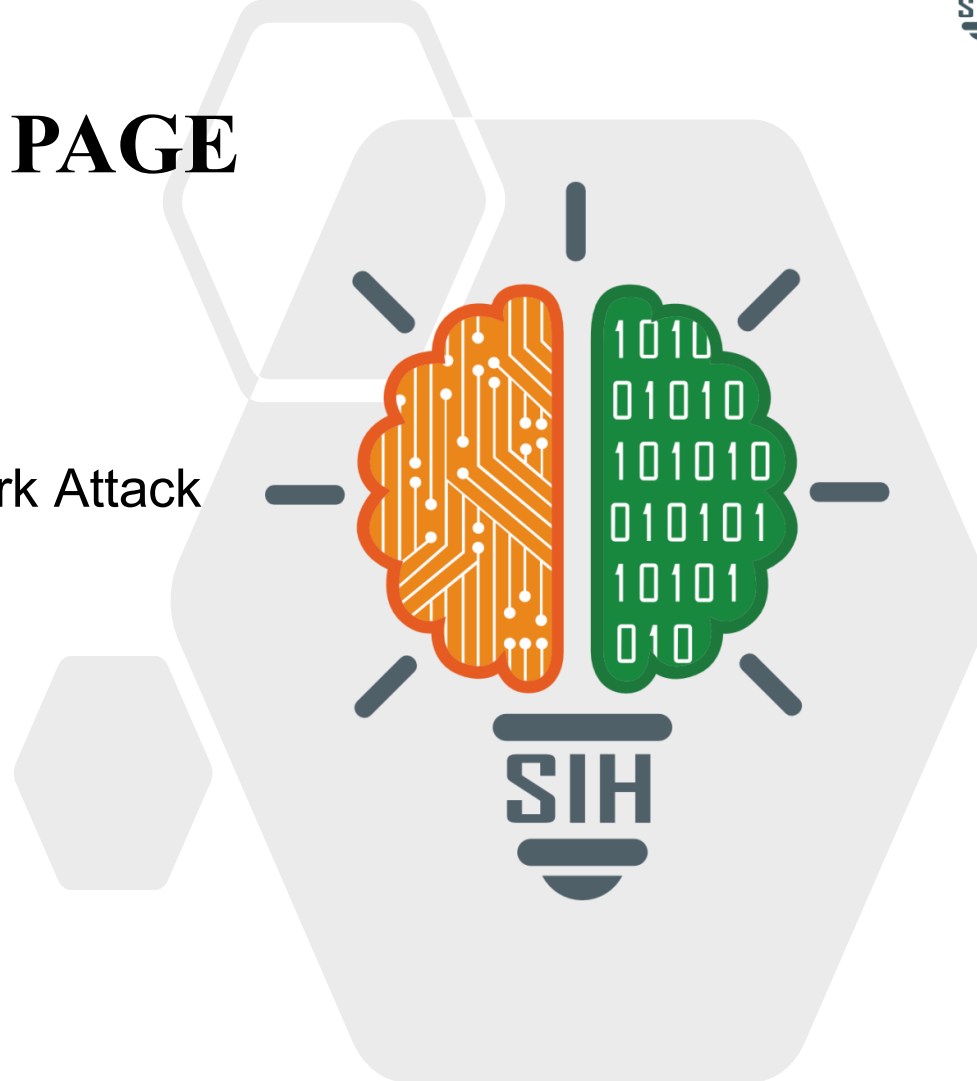


TITLE PAGE

- **Problem Statement ID** – SIH26153
- **Problem Statement Title**- AI based Network Attack
Forecasting from Network Traffic Data
- **Theme**- Blockchain & Cybersecurity
- **PS Category**- Software
- **Team ID**- 127364
- **Team Name (Registered on portal)**- VIGHNAX



VIGHNAX: a world model that forecasts network attacks before they complete



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

❖ Detailed explanation of the proposed solution

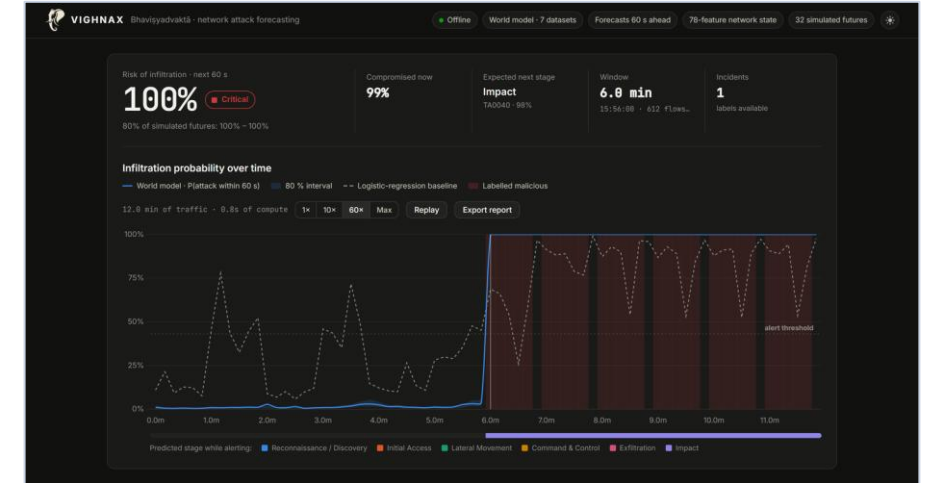
- **A world model of the network:** every 10 s it reads a 78-feature state from flow records and/or packets
- **Simulates 32 futures, 60 s ahead:** attack probability with an uncertainty band, and the next MITRE ATT&CK stage
- **Explains every alert:** Integrated Gradients, plus the hosts and flows responsible

❖ How it addresses the problem

- **Early warning, not after-the-fact detection:** 54 of 201 unseen attack onsets forecast before they began (baseline: 17)
- **One model, all seven PS datasets:** tested only on days, captures and botnets it never saw

❖ Innovation and uniqueness of the solution

- **'What-if' simulation** of the whole network, not a per-flow classifier
- **Answers who, not only what:** re-forecasting without each host ranks the hosts behind an alert
- **Air-gap ready:** laptop CPU, SHA-256-verified weights, one-click incident report



Working prototype on an unseen CIC-IDS2017 day: risk turns critical as the DDoS begins.

54/201

attack onsets
warned before
they began

3.2×

more early
warnings than
the baseline (17)

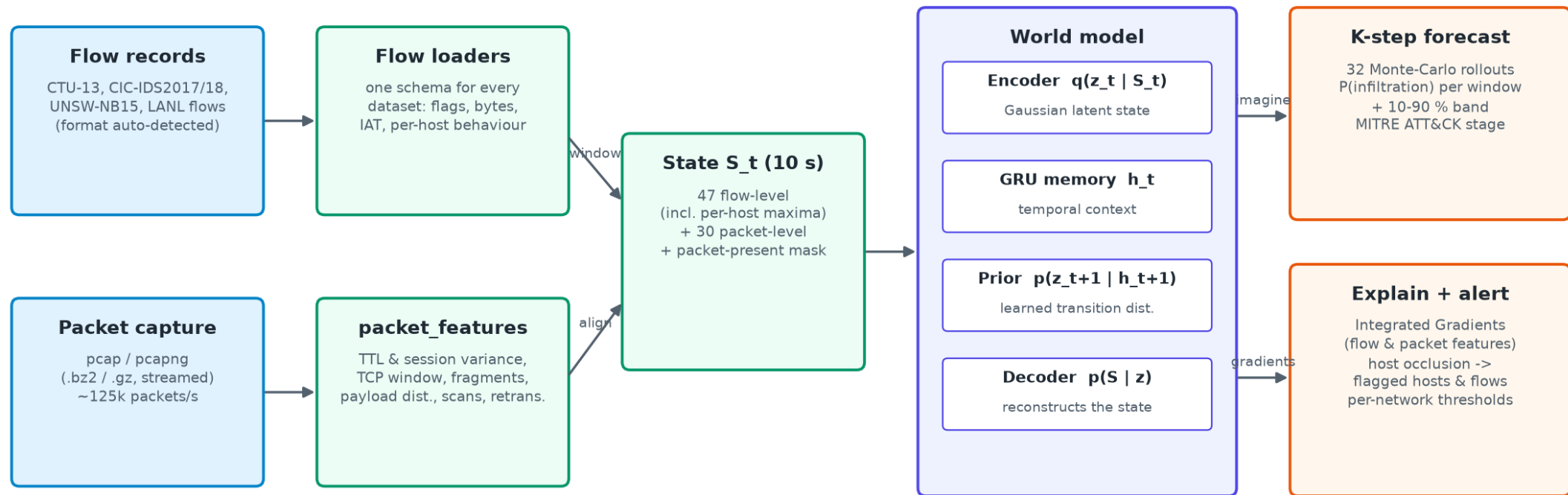
~8 ms

per 10 s window
on a laptop CPU

❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

- **Model:** Python, PyTorch (world model), scikit-learn (baseline); dpkt / Scapy for PCAP, pandas / NumPy
- **Interface:** Flask + Chart.js dashboard with bundled fonts and scripts, no internet or cloud API
- **Hardware:** any laptop CPU (~8 ms per 10 s window); a GPU only speeds up training

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)



Offline dashboard: live timeline, forecast band, stage mix, attributions, flagged hosts & flows, incident report | no cloud APIs

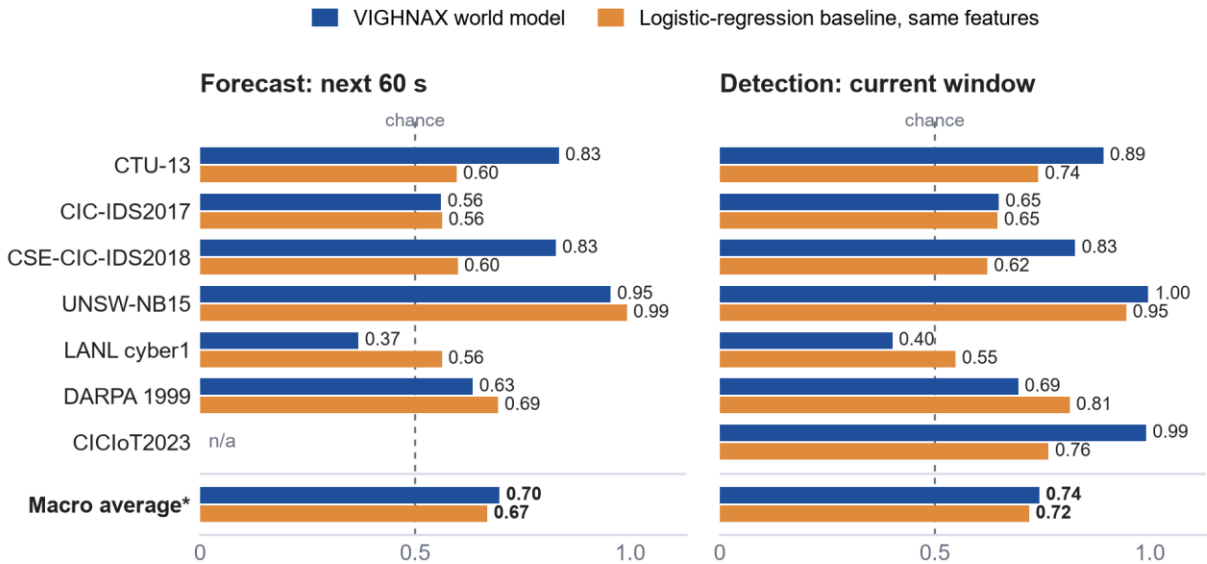


FEASIBILITY AND VIABILITY



❖ Analysis of the feasibility of the idea

- **Already built:** working prototype, trained weights and the full benchmark are public on GitHub
- **Tested on unseen data:** macro forecast AUROC 0.70 vs 0.67 for the baseline, detection 0.74 vs 0.72
- **Fast and light:** one laptop CPU core processes traffic over $1,000\times$ faster than real time
- **Easy to deploy:** one-click launch; PCAP or flow CSV in; no GPU, cloud or internet needed
- **Honest evaluation:** scored only on traffic it never saw; weaker results are shown, not hidden



AUROC on unseen test data; 0.5 = chance. CICIoT2023 episodes are composed, so only detection is scored.
*Mean over the six temporal datasets.

Potential challenges and risks

Some attack types: weak on LANL (authentication-based red team), behind the baseline on DARPA 1999

New networks: more false alarms where the model was not calibrated; behaviour drifts over time

Encryption: payloads are hidden in encrypted traffic

Strategies for overcoming these challenges

Authentication-log and per-host modelling; these limits are reported openly, not hidden

Per-network alert calibration on the site's own traffic (built in); periodic retraining

Flow-level features need no payload, so the model works on encrypted and flow-only input

❖ Potential impact on the target audience

- **Early warning** for NTRO, NCIIPC, CERT-In and critical-infrastructure SOCs, with time to isolate hosts
- **Faster triage:** every alert comes with its reasons and the hosts and flows behind it
- **Deployable anywhere:** air-gapped, on ordinary hardware, including classified networks

❖ Benefits of the solution (social, economic, environmental, etc.)

Security

Earlier containment of botnets, DDoS and lateral movement on national infrastructure

Social

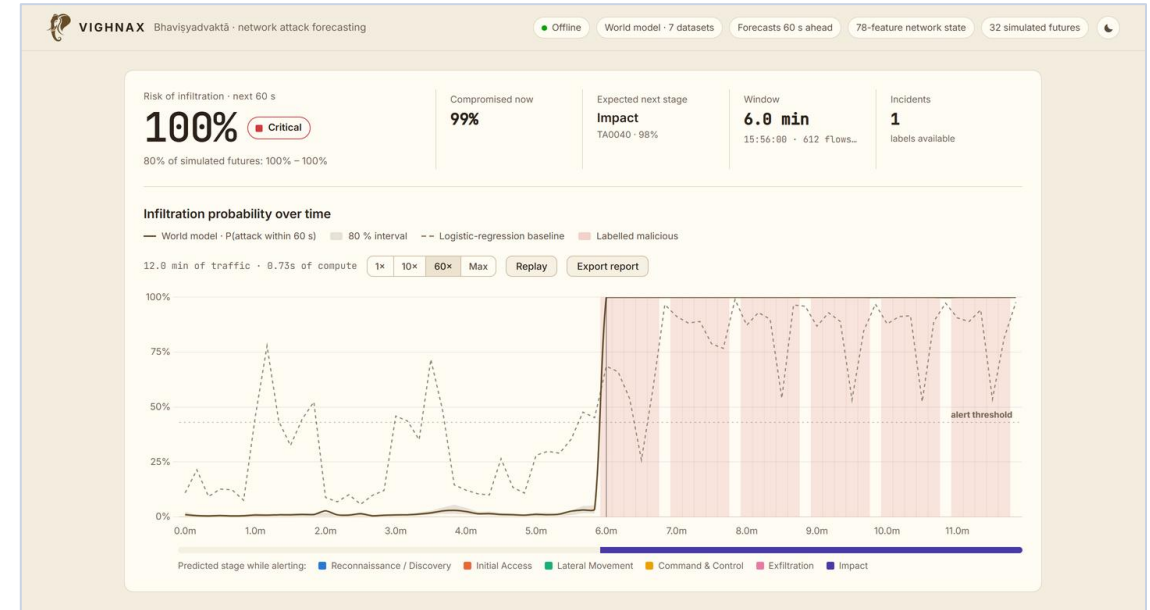
Protects power, banking, health and e-governance services that citizens use

Economic

Open source: no licence or cloud cost; one CPU core per network segment

Sovereignty

Indigenous, auditable AI; verified weights; no data leaves the network



A real CIC-IDS2017 day never seen in training: risk stays low for six minutes, then goes critical as the DDoS starts; the baseline (dashed) raises false spikes.



Watch the 2-minute demo

https://youtu.be/pPTe-d_TByU

Scan the code or click the link

❖ Details / Links of the reference and research work

Research

- **World models:** Ha & Schmidhuber, World Models (2018); Hafner et al., recurrent state-space model, Learning Latent Dynamics for Planning from Pixels, ICML 2019
- **Explainability:** Sundararajan, Taly & Yan, Axiomatic Attribution for Deep Networks (Integrated Gradients), ICML 2017
- **Attack stages and guidance:** MITRE ATT&CK Enterprise matrix (attack.mitre.org); NCIIPC (nciipc.gov.in)

Datasets: all seven named in the PS

- CTU-13 (Garcia et al., 2014) · CIC-IDS2017 (Sharafaldin et al., 2018) · CSE-CIC-IDS2018 (CIC & AWS, 2018) · UNSW-NB15 (Moustafa & Slay, 2015)
- LANL cyber1 (Kent, 2015) · DARPA 1999 (Lippmann et al., 2000) · CICIOT2023 (Neto et al., 2023)

Our work

- **Source code, weights, benchmark:** <https://github.com/NAMAN-THAKUR-1944/SIH2026-SIH26153-VIGHNAX>
- **Demo video (2 min):** https://youtu.be/pPTe-d_TByU
- **Reproducible:** one configuration (configs/default.yaml, seed 42); python evaluate.py regenerates every number in this deck from the public datasets



Source code

GitHub · weights · benchmark



Demo video

YouTube · 2 minutes